

CSA1613 (Data Warehouse and
Data Mining)

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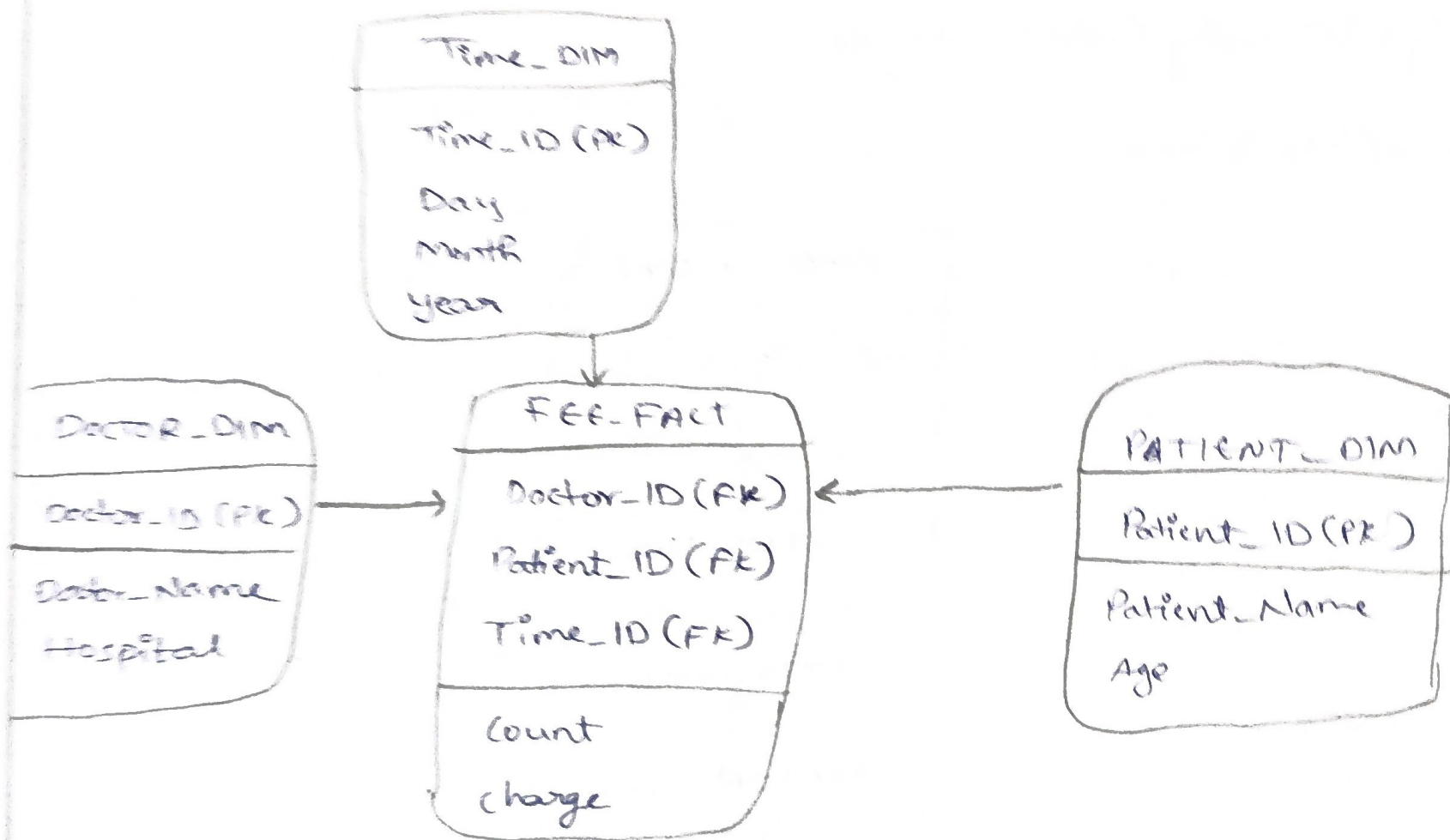
Short Answer Test (Set - 2) :

Data - Fact Data Warehouse:

Q1 Time Series Data warehouse schema:

- * Star Schema
- * Snowflake Schema
- * Galaxy (Fact constellation) schema

Q2 Star Schema:



Fact Table: (Fee-Fact)

measures:

is Count
is Charge

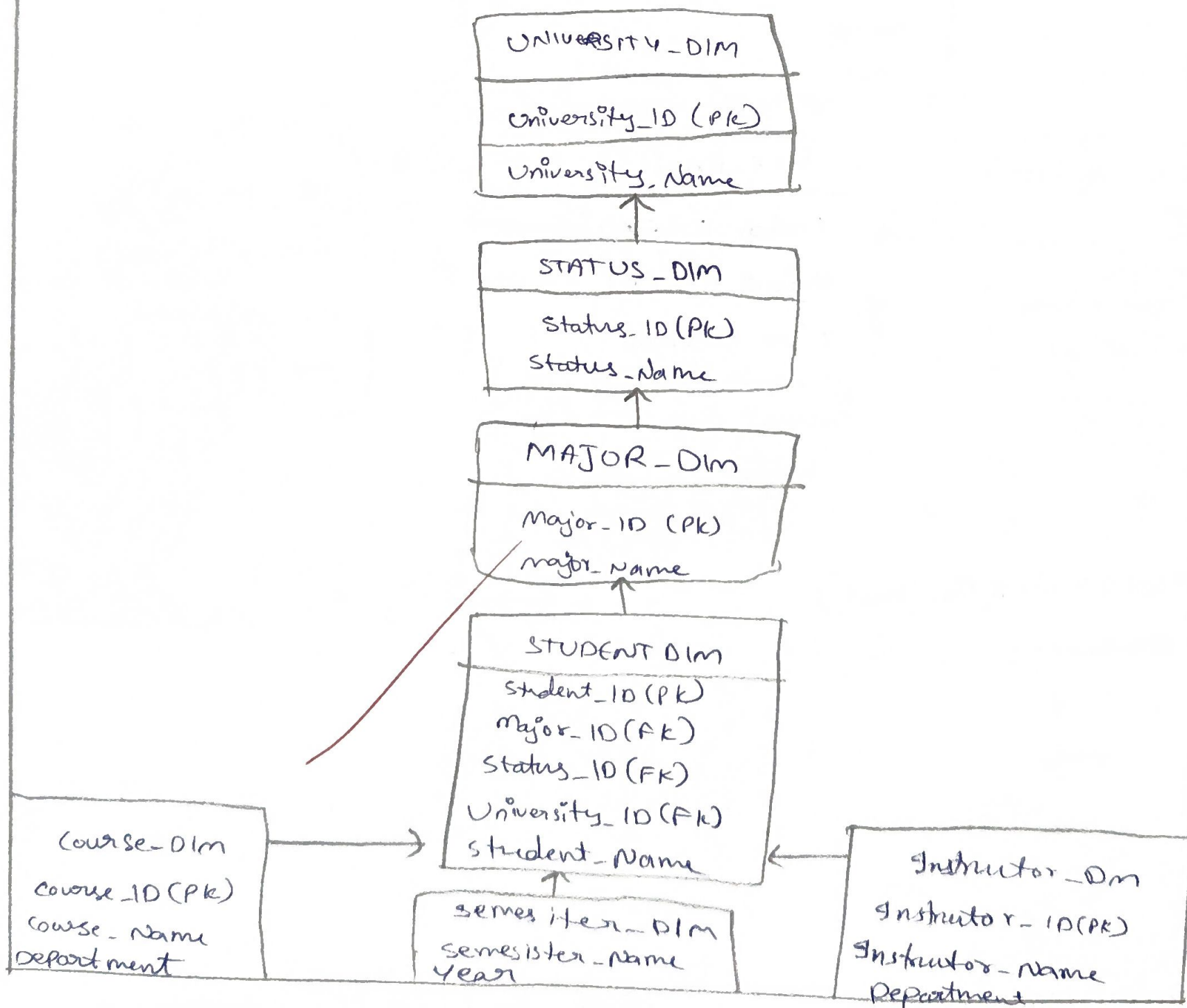


(c.) SQL Query :-

```
SELECT
    doctor,
    year,
    SUM(count) AS Total-visits,
    AVG(charge) AS Average-charge
FROM Fee
GROUP BY doctor, year;
```

2. Big University Data Warehouse :-

(a) Snowflake Schema :-



4. (b)

OLAP Operations:

Base Cuboid

[Student, course, Semester, Instructor]

Operations:-

1. Roll-Up Semester $\rightarrow$ Year
2. Slice Course = CS
- 3 Roll-Up Instructor $\rightarrow$ ALL
4. Keep Student

Final Cuboid

[Student, CS course, Year, ALL Instructor]

Output:-

Average Grade of CS Courses for each student.

(c) Number of Cuboids:-

Each dimension has 5 levels.

$$\begin{aligned}\text{Formula - Cuboids} &= 5 \times 5 \times 5 \times 5 \\ &= 5^4 \\ &= 625\end{aligned}$$

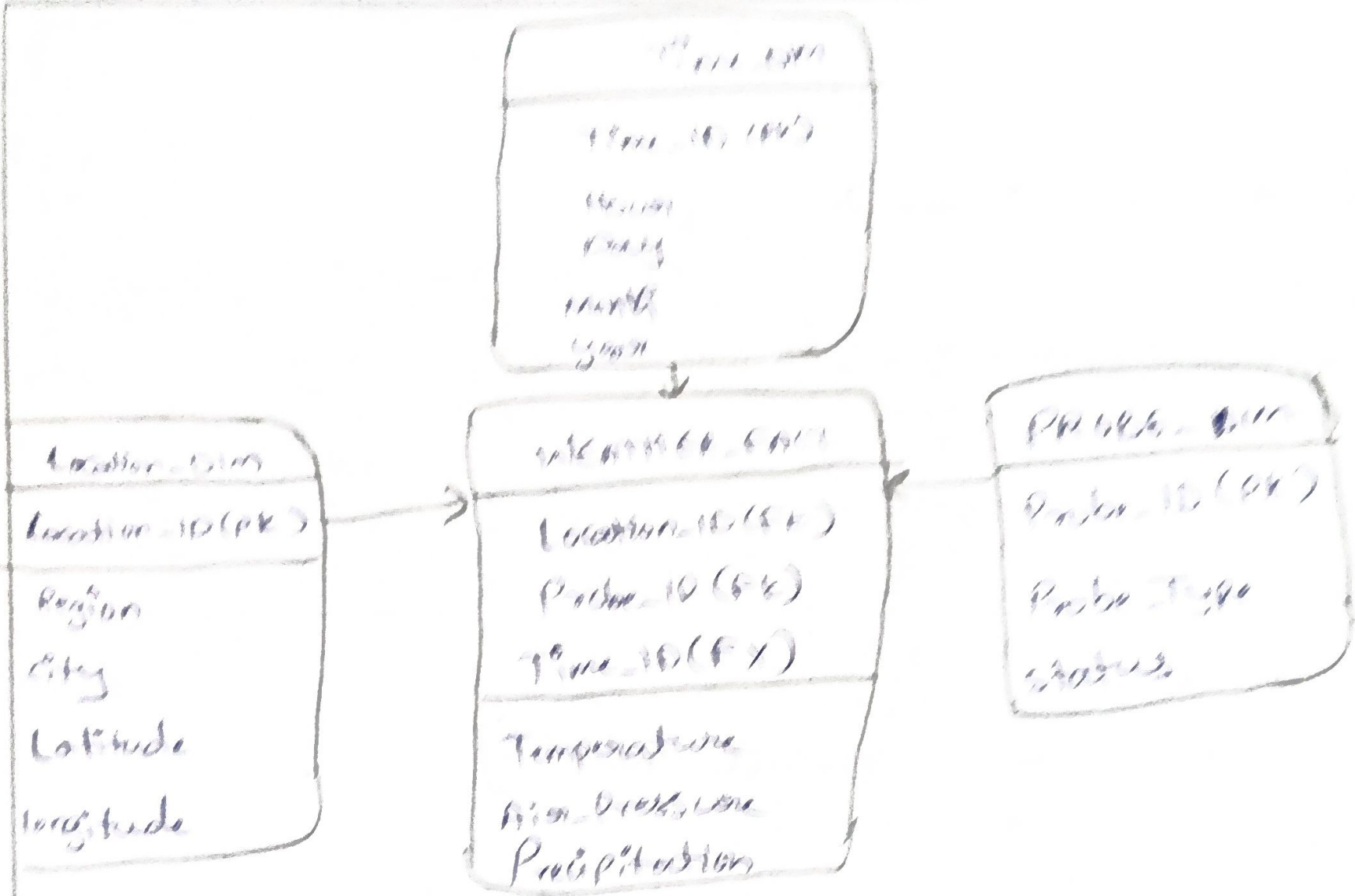
$\therefore$ 625 cuboids.

7.

Regional Weather Bureau Data Warehouse:-

Dimensions:-

- $\rightarrow$ Time
- $\rightarrow$ Location
- $\rightarrow$ Probe



Measures:

- * Temperature
- * Air Pressure
- * Precipitation

Benefits:

- Analyse weather trends.
- Compare temperatures across regions.
- Study seasonal weather changes.
- Support risk and analysis.
- Predict future weather patterns.

4. Data cube :-
Given,
n dimensions
p values in each dimension

(a) max cells in Base cuboid :-

$$\text{Formula} = p^n$$

(b) Min cells in Base cuboid :-

If every dimension has only one existing combination

Ans - 1.

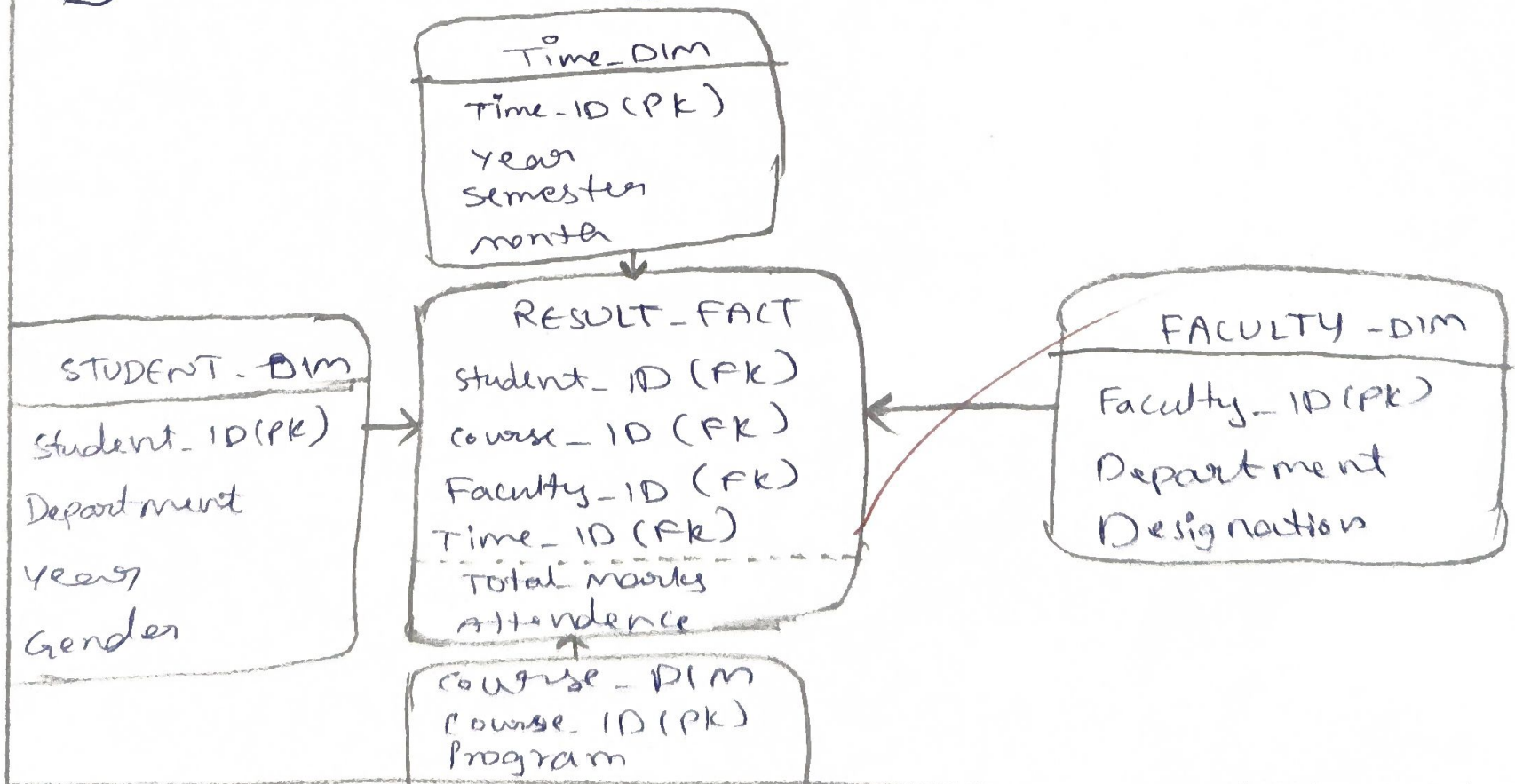
(c) Max cells in Entire cube :-

$$(P+1)^n = \text{Formula}$$

(d) min cells in Entire cube :-

$$\text{Formula} = 2^n$$

5. University Academic Performance Data warehouse :-



b) OLAP Operations:-

Base Cuboid

[Student, Course, Faculty, Time]

Operations:-

1. Roll-up Time $\rightarrow$ Academic Year
2. Roll-up student $\rightarrow$ Department
3. Roll-up Faculty $\rightarrow$ All
4. Aggregate Average Marks

Final Result:-

Department-wise Average marks for each Academic Year

c) No. of Cuboids:-

Given,

Student = 4 levels

Course = 3 levels

Faculty = 3 levels

Time = 5 levels

$$\text{Formula} = 4 \times 3 \times 3 \times 5 \\ = 180$$

$\therefore 180$ Cuboids.

